

What we're doing

Sourcing candidate eval structures that are deliberately ****far from** MarFold's AFDB training set**, for issue #65 (continuation of #41). The #41 Foldseek train-similarity tool showed FoldBench-100 is almost entirely close to training (48 redundant / 51 same-fold / 1 novel). To actually probe generalization we need candidates from the novel / shallow-MSA regime, which a depth-agnostic PDB sample barely contains.

Why these three sources

A single-sequence model has the most to prove where MSA-based models lose their coevolution signal (shallow MSA) and where the fold is novel. Three sources hit that, each for a different reason:

- **PDB de novo proteins** — no evolutionary lineage ($N_{eff} \sim 1$), often novel folds, mostly outside UniProt/AFDB.
- **CASP14/15 free-modeling domains** — the community "no usable template, few homologs" gold standard; blind and temporally honest.
- **CAMEO hard targets** — a rolling weekly stream of difficult monomers.

What we fetched

Three idempotent fetch scripts sharing one manifest schema (so the sources stack and join onto exp12/exp41 CSVs):

- de novo: RCSB search → **396** experimental monomers (40-400 aa, res ≤ 3.0) out of 2,007 in the DE NOVO PROTEIN class.
- CASP FM: **32** FM/FM-TBM domains (17 CASP14 + 15 CASP15); **26** resolved (19 from the public domain tarballs + 7 clipped out of the deposited PDB entry for oligomeric / late-release targets), 6 with no released structure recorded without coords (not silently dropped).
- CAMEO hard: **35** hard targets over the latest 1-year window, references pulled from RCSB by PDB id.

Fold novelty vs the training set

Foldseek-TM each candidate vs the 1.33M AFDB-24M train reps (exp41 tool). **Novel folds: CASP FM 50% / CAMEO hard 34% / de novo 6%** — versus FoldBench-100's 1/100. De novo designs cluster near known folds structurally but are sequence-novel (only 41% have a $\geq 30\%$ -id training-sequence match).

MSA depth (the named axis)

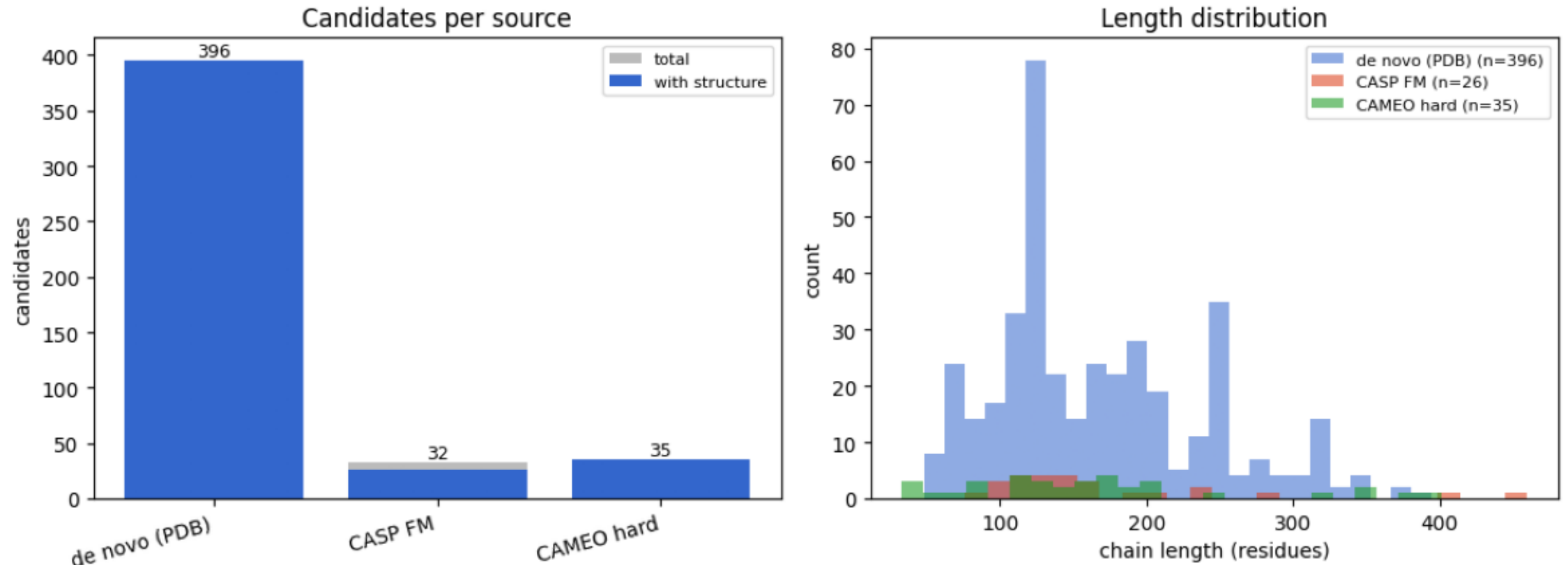
ColabFold MMseqs2 MSA per candidate (no Modal; the search is on the free ColabFold server either way) -> Neff. **221/454 (49%) have Neff < 10** — the shallow-MSA regime where a single-sequence model has the most headroom. The assessor-released CASP FM monomers are fold-novel but homolog-rich (deep MSA); the oligomeric / late-release FM targets recovered from the PDB are mostly orphan-MSA.

The headline: fold novelty x MSA depth

The 2-D label populates the cell FoldBench could not: **33 candidates are novel-fold AND shallow-MSA** (21 de novo, 6 CAMEO, 6 CASP FM). The per-protein table (``data/candidate_2d_label.csv``) carries all three axes (fold novelty, sequence leakage, MSA depth) so the eval can stratify MarFold vs MSA baselines. Next: accuracy on these strata.

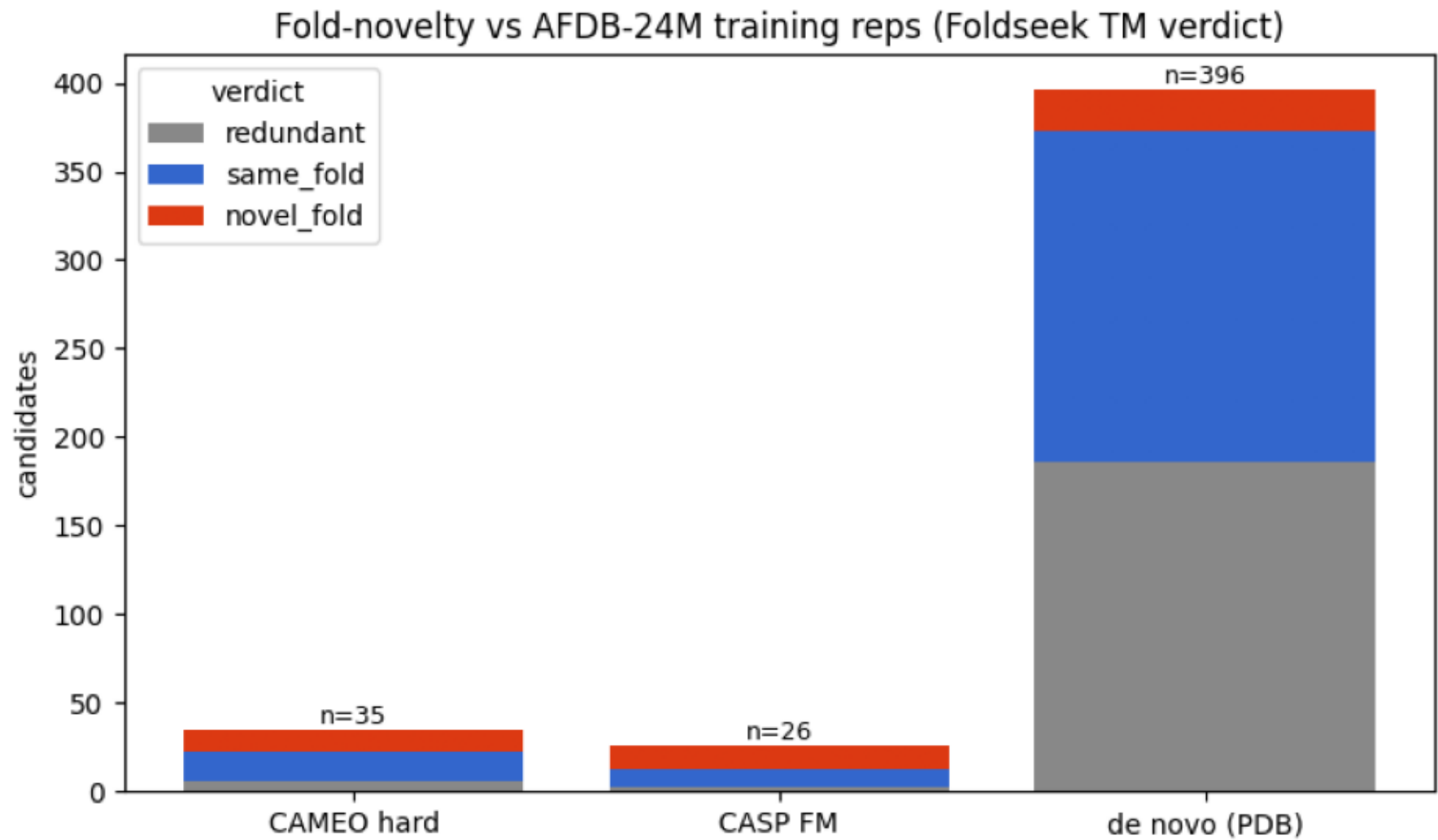
candidates_overview.png

Fetches exp65 candidate set. Left: total vs with-downloaded-structure counts per source (CASP FM's gap = oligomeric/late-release domains recorded without coords). Right: chain-length distributions.



novelty_by_source.png

Fold-novelty of each candidate source vs the 1.33M AFDB-24M training representatives (exp41 Foldseek query; qtm score verdict, redundant $\geq$ 0.9 / same_fold $\geq$ 0.5 / novel_fold). Compare to FoldBench-100's 1/100 novel.



two_axis_label.png

exp65 2-D eval label. Left: fold-novelty (vs AFDB-24M train reps, Foldseek) x MSA-depth (Neff, ColabFold); red box = the novel-fold x shallow-MSA corner FoldBench-100 couldn't populate. Right: which datasets supply the shallow-MSA tail.

